Safety Data Sheet

acc. to 29 CFR 1910.1200 App D

LIFT OFF

Revision Date: 12/08/2025

SECTION 1: Identification

1.1 Product Identifier

Trade Name	**LIFT OFF**
Product Number	10-10085

1.2 Relevant Identified Uses

Relevant Identified Uses	Acidic Presoak

1.3 Details of the Supplier of the Safety Data Sheet

ChemQuest Inc.
21365 Hamburg Avenue
Lakeville, MN 55044
United States

Telephone: +1 (952) 985-9993
e-mail: regulatory@carwashsoap.com
Website: https://carwashsoap.com/

1.4 Emergency Telephone Number

Emergency Information Service	ChemTrec 1-800-424-9300

SECTION 2: Hazard(s) Identification

2.1 Classification of the Substance

Classification Acc. to OSHA "Hazard Communication Standard" (29 CFR 1910.1200)

Hazard Statement	Hazard Class	Category
H314	skin corrosion/irritation	1B
H318	serious eye damage/eye irritation	1
H350	carcinogenicity	1A

Employ good industrial hygiene practice

2.2 Label Elements

Labelling acc. to OSHA "Hazard Communication Standard" (29 CFR 1910.1200)

Signal Word **DANGER**

- Hazard Statements

H314	Causes severe skin burns and eye damage.
H350	May cause cancer.

- Precautionary Statements

P201	Obtain special instructions before use.
P260	Do not breathe dusts or mists.
P280	Wear eye protection/face protection.
P301+P330+P331	If swallowed: Rinse mouth. Do NOT induce vomiting.
P303+P361+P353	If on skin (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower.
P304+P340	If inhaled: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338	If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a poison center/doctor.
P321	Specific treatment (see on this label).
P363	Wash contaminated clothing before reuse.
P405	Store locked up.
P501	Dispose of contents/container to industrial combustion plant.

2.3 Other Hazards

Hazards Not Otherwise Classified

Harmful to aquatic life with long lasting effects (GHS category 3: aquatic toxicity - acute and/or chronic).

SECTION 3: Composition/Information on Ingredients

3.1 Substances

Not applicable

3.2 Mixtures

Description of the Mixture

Name of Substance	CAS No	Wt%
Water	7732-18-5	50 – < 75
Sulfuric acid	7664-93-9	5 – < 10
2-Butoxyethanol	111-76-2	5 – < 10
Phosphoric acid	7664-38-2	5 – < 10
Dodecylbenzenesulfonic acid	68584-22-5 27176-87-0	1 – < 5
Sulfonic acids, C14-16 alkane hydroxy and C14-16 alkene, sodium salts	68439-57-6	1 – < 5
Sodium Xylenesulfonate	1300-72-7	1 – < 5
Cocoamidopropyl hydroxysultaine	68139-30-0	1 – < 5
Proprietary Surfactant	Proprietary	1 – < 5

SECTION 4: First-Aid Measures

4.1 Description of First-Aid Measures

General Notes

Do not leave affected person unattended. Remove victim out of the danger area. Take off immediately all contaminated clothing. In all cases of doubt, or when symptoms persist, seek medical advice. In case of unconsciousness place person in the recovery position. Never give anything by mouth.

Following Inhalation

Provide fresh air. If breathing is irregular or stopped, immediately seek medical assistance and start first aid actions. In case of respiratory tract irritation, consult a physician.

Following Skin Contact

Wash with plenty of soap and water.

Following Eye Contact

Remove contact lenses, if present and easy to do. Continue rinsing. Irrigate copiously with clean, fresh water for at least 15 minutes, holding the eyelids apart.

Following Ingestion

Rinse mouth with water (only if the person is conscious). Do NOT induce vomiting.

4.2 Most Important Symptoms and Effects, Both Acute and Delayed

Symptoms and effects are not known to date.

SECTION 5: Fire-Fighting Measures

5.1 Extinguishing Media

Suitable Extinguishing Media

Water spray, Alcohol resistant foam, BC-powder, Carbon dioxide (CO2)

5.2 Special Hazards Arising from the Substance or Mixture

Contact with metals may emit flammable hydrogen gas.

5.3 Fire-Fighting Measures

In case of fire and/or explosion do not breathe fumes. Coordinate firefighting measures to the fire surroundings. Do not allow firefighting water to enter drains or water courses. Collect contaminated firefighting water separately. Fight fire with normal precautions from a reasonable distance.

SECTION 6: Accidental Release Measures

6.1 Personal Precautions, Protective Equipment and Emergency Procedures

For Non-Emergency Personnel

Remove persons to safety.

For Emergency Responders

Wear breathing apparatus if exposed to vapors/dust/aerosols/gases.

6.2 Environmental Precautions

Do not empty into drains, surface water or soil. If the product has entered a water course, sewer or soil, inform the responsible authority.

6.3 Methods and Material for Containment and Cleaning Up

Advice on How to Contain a Spill

Prevent entry to sewers and public waters. Contain any spills with dikes or absorbents to prevent migration and entry into sewers or streams.

Advice on How to Clean Up a Spill

Collect spillage. Ensure good ventilation and exhaustion. Place in appropriate containers for disposal.

6.4 Reference to Other Sections

Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

SECTION 7: Handling and Storage

7.1 Precautions for Safe Handling

Measures to Prevent Fire as well as Aerosol and Dust Generation

Use local and general ventilation. Use only in well-ventilated areas.

- Handling of Incompatible Substances or Mixtures

- Keep Away From

Caustic solutions

Advice on General Occupational Hygiene

Wash hands after use. Do not eat, drink and smoke in work areas. Remove contaminated clothing and protective equipment before entering eating areas. Never keep food or drink in the vicinity of chemicals. Never place chemicals

in containers that are normally used for food or drink. Keep away from food, drink and animal feedingstuffs.

7.2 Conditions for Safe Storage, Including Any Incompatibilities

Packaging Compatibilities

Only packagings which are approved (e.g. acc. to the Dangerous Goods Regulations) may be used.

SECTION 8: Exposure Controls/Personal Protection

8.1 Control Parameters

Occupational Exposure Limit Values (Workplace Exposure Limits)

Country	Name of Agent	CAS No	Identifier	TWA [ppm]	TWA [mg/m³]	STEL [ppm]	STEL [mg/m³]	Ceiling-C [ppm]	Ceiling-C [mg/m³]	Notation	Source
US	2-butoxyethanol	111-76-2	TLV®	20							ACGIH® 2024
US	2-butoxyethanol	111-76-2	REL	5 (10 h)	24 (10 h)					H	NIOSH REL
US	2-butoxyethanol	111-76-2	PEL	50	240					H	29 CFR 1910.1000
US	2-butoxyethanol (EGBE) (glycol monobutyl ether)	111-76-2	PEL (CA)	20	97					H	Cal/OSHA PEL
US	phosphoric acid	7664-38-2	PEL (CA)		1		3				Cal/OSHA PEL
US	phosphoric acid	7664-38-2	REL		1 (10 h)		3				NIOSH REL
US	phosphoric acid	7664-38-2	TLV®		1		3				ACGIH® 2024
US	phosphoric acid	7664-38-2	PEL		1						29 CFR 1910.1000
US	sulfuric acid	7664-93-9	PEL (CA)		0.1		3				Cal/OSHA PEL
US	sulfuric acid	7664-93-9	REL		1 (10 h)						NIOSH REL
US	sulfuric acid	7664-93-9	PEL		1						29 CFR 1910.1000
US	sulfuric acid	7664-93-9	TLV®		0.2					t	ACGIH® 2024

Notation

Ceiling-C	ceiling value is a limit value above which exposure should not occur
H	absorbed through the skin
STEL	short-term exposure limit: a limit value above which exposure should not occur and which is related to a 15-minute period (unless otherwise specified)
t	thoracic fraction
TWA	time-weighted average (long-term exposure limit): measured or calculated in relation to a reference period of 8 hours time-weighted average (unless otherwise specified

Biological limit values

Country	Name of Agent	Parameter	Nota-tion	Identifier	Value	Source
US	2-butoxyethanol	Butoxyacetic acid (BAA)	hydr, crea, crea-lim-it3c	BEI®	200 mg/g	ACGIH® 2024

Notation

crea	creatinine
crea-limit3c	acceptable limits on urine specimens: creatinine concentration: >0.3 g/L and <3.0 g/L, specific gravity:>1.010 and <1.030
hydr	hydrolysis

8.2 Exposure Controls

Appropriate Engineering Controls

General ventilation.

Individual Protection Measures (Personal Protective Equipment)

Eye/Face Protection

Wear eye/face protection.

Skin Protection

- Hand Protection

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. Check leak-tightness/impermeability prior to use. In the case of wanting to use the gloves again, clean them before taking off and air them well. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.

- Body Protection

Wear suitable protective clothing. Wear suitable face shield.

- Other Protection Measures

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended. Wash hands thoroughly after handling.

Respiratory Protection

In case of inadequate ventilation wear respiratory protection.

Environmental Exposure Controls

Use appropriate container to avoid environmental contamination. Keep away from drains, surface and ground water.

SECTION 9: Physical and Chemical Properties

9.1 Information on Basic Physical and Chemical Properties

Appearance

Physical State	Liquid
Color	Straw
Foam Color	White

Fragrance	None

Other safety parameters

pH (value)	1.5 (acid)
Melting Point/Freezing Point	No Data Available
Initial boiling point and boiling range	No Data Available

Flash Point	No Data Available
Evaporation rate	No Data Available
Flammability (solid, gas)	No Data Available
Vapor pressure	No Data Available
Density	1.094 $^{g}/_{ml}$

Solubility(ies)

- Water solubility	Miscible in Any Proportion

Viscosity

- Kinematic viscosity	No Data Available
Oxidizing Properties	None
9.2 Other Information	There Is No Additional Information

SECTION 10: Stability and Reactivity

10.1 Reactivity
Concerning incompatibility: see below "Conditions to avoid" and "Incompatible materials".

10.2 Chemical Stability
See below "Conditions to avoid".

10.3 Possibility of Hazardous Reactions
No known hazardous reactions.

10.4 Conditions to Avoid
There are no specific conditions known which have to be avoided.

10.5 Incompatible Materials
Bases, Strong Oxidizers, Reducing Agents, Metals

Release of flammable materials with:
Light metals (due to the release of hydrogen in an acid/alkaline medium)

10.6 Hazardous Decomposition Products
Hazardous combustion products: see section 5.

SECTION 11: Toxicological Information

11.1 Information on Toxicological Effects
Test data are not available for the complete mixture.

Classification Procedure
The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

Classification Acc. to OSHA "Hazard Communication Standard" (29 CFR 1910.1200)

Acute Toxicity
Shall not be classified as acutely toxic.

Skin Corrosion/Irritation
Causes severe skin burns and eye damage.

Serious eye damage/eye irritation
Causes serious eye damage.

Respiratory or Skin Sensitization

Shall not be classified as a respiratory or skin sensitizer.

Germ Cell Mutagenicity

Shall not be classified as germ cell mutagenic.

Carcinogenicity

May cause cancer.

IARC Monographs on the Evaluation of Carcinogenic Risks to Humans

Name of Substance	CAS No	Classification	Number
Sulfuric acid	7664-93-9	1	
2-butoxyethanol	111-76-2	3	

Legend

1	Carcinogenic to humans
3	Not classifiable as to carcinogenicity in humans

National Toxicology Program (United States): Report on Carcinogens

Name of Substance	CAS No	Classification	Number
Sulfuric acid	7664-93-9	Known to be a human carcinogen	9th Report on Carcinogens

Reproductive Toxicity

Shall not be classified as a reproductive toxicant.

Specific Target Organ Toxicity - Single Exposure

Shall not be classified as a specific target organ toxicant (single exposure).

Specific Target Organ Toxicity - Repeated Exposure

Shall not be classified as a specific target organ toxicant (repeated exposure).

Aspiration Hazard

Shall not be classified as presenting an aspiration hazard.

SECTION 12: Ecological Information

12.1 Toxicity

Harmful to aquatic life with long lasting effects.

12.2 Persistence and Degradability

No Data Available.

12.3 Bioaccumulative Potential

No Data Available.

12.4 Mobility in Soil

No Data Available.

12.5 Other Adverse Effects

No Data Available.

SECTION 13: Disposal Considerations

13.1 Waste Treatment Methods

Sewage Disposal-Relevant Information

Do not empty into drains. Avoid release to the environment.

Waste Treatment of Containers/Packages

Only packagings which are approved (e.g. acc. to DOT) may be used. Completely emptied packages can be recycled. Handle contaminated packages in the same way as the substance itself.

Remarks

Please consider the relevant national or regional provisions. Waste shall be separated into the categories that can be handled separately by the local or national waste management facilities.

SECTION 14: Transport Information

14.1 UN Number, Proper Shipping Name, Class and Packing Group

Domestic Ground Non-Bulk Shipments

UN1760, CORROSIVE LIQUID, N.O.S., (SULFURIC ACID / PHOSPHORIC ACID) 8, II

14.2 Special precautions for user

There Is No Additional Information.

Transport of dangerous goods by road or rail (49 CFR US DOT) - Additional information

Reportable quantity (RQ) of LIFT OFF 13,930 lbs (Sulfuric acid) (phosphoric acid)

SECTION 15: Regulatory Information

15.1 Safety, Health and Environmental Regulations Specific for the Product in Question

National Regulations (United States)

Toxic Substance Control Act (TSCA) all ingredients are listed

Superfund Amendment and Reauthorization Act (SARA TITLE III)

- The List of Extremely Hazardous Substances and Their Threshold Planning Quantities (EPCRA Section 302, 304)

The List of Extremely Hazardous Substances and Their Threshold Planning Quantities

Name of Substance	CAS No	Notes	Reportable quantity (pounds)	Threshold planning quantity (pounds)
Sulfuric acid	7664-93-9		1,000	1000

- Specific Toxic Chemical Listings (EPCRA Section 313)

Toxics Release Inventory: Specific Toxic Chemical Listings

Name of Substance	CAS No	Remarks	Effective date
Sulfuric acid	7664-93-9	acid aerosols including mists, vapors, gas, fog, and other airborne forms of any particle size	12/31/1986

Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA)

- List of Hazardous Substances and Reportable Quantities (CERCLA section 102a) (40 CFR 302.4)

Name of Substance	CAS No	Final RQ pounds (Kg)
Sulfuric acid	7664-93-9	1000 (454)
Dodecylbenzenesulfonic acid	27176-87-0	1000 (454)
phosphoric acid	7664-38-2	5000 (2270)

California Environmental Protection Agency (Cal/EPA): Proposition 65 - Safe Drinking Water and Toxic Enforcement Act of 1987

Proposition 65 List of chemicals

Name of Substance	Name acc. to inventory	CAS No	Type of the toxicity	Date listed
1,4-dioxane	1,4-dioxane	123-91-1	cancer	12/31/1987
Sulfur dioxide	sulfur dioxide	7446-09-5	developmental	07/28/2011
Ethane-1,2-diol	ethylene glycol (ethanediol)	107-21-1	developmental	06/18/2015
Cumene	cumene	98-82-8	cancer	04/05/2010
Benzene	benzene	71-43-2	cancer	02/26/1987
Benzene	benzene	71-43-2	developmental, male	12/25/1997
Toluene	toluene	108-88-3	developmental	12/31/1990
Ethylbenzene	ethylbenzene	100-41-4	cancer	06/10/2004

Right to Know Hazardous Substance List

- Cleaning Product Right to Know Act Substance List (CA-RTK)

Name of Substance	CAS No	Functionality	Authoritative Lists
Sulfuric acid	7664-93-9		IARC Carcinogens - 1 NTP 13th RoC - known OEHHA RELs Prop 65
2-butoxyethanol	111-76-2	solvents	ATSDR Neurotoxicants OEHHA RELs
2-butoxyethanol			CA TACs
phosphoric acid	7664-38-2		OEHHA RELs

- Toxic or Hazardous Substance List (MA-TURA)

Name of Substance	CAS No	DEP CODE	PBT / HHS / LHS	PBT / HHS Threshold	De Minimis Concentration Threshold
Sulfuric acid	7664-93-9				1.0 %
Dodecylbenzenesulfonic acid	27176-87-0				1.0 %
2-butoxyethanol		1022			1.0 %
phosphoric acid	7664-38-2				1.0 %

- Hazardous Substances List (MN-ERTK)

Name of Substance	CAS No	References	Remarks
Sulfuric acid	7664-93-9	A, N, O	
2-butoxyethanol	111-76-2	A, O	skin
phosphoric acid	7664-38-2	A, O	

Legend

A	American Conference of Governmental Industrial Hygienists (ACGIH), "Threshold Limit Values for Chemical Substances and Physical Agents and Biological Exposure Indices for 1992-93", available from ACGIH
N	National Institute for Occupational Safety and Health (NIOSH), "Recommendations for Occupational Safety and Health Standards," August 1988, available from NIOSH, Publications Dissemination Office, Division of Standards Development and Technology Transfer
O	Occupational Safety and Health Administration (OSHA), Safety and Health Standards, Code of Federal Regulations, title 29, part 1910, subpart Z, "Toxic and Hazardous Substances, 1990." General information: Minnesota Department of Labor and Industry, Occupational Safety and Health Division
skin	If a potential for absorption from skin contact merits special consideration, the word "skin" follows the substance name.

- Hazardous Substance List (NJ-RTK)

Name of Substance	CAS No	Remarks	Classifications
Sulfuric acid	7664-93-9		CA CO R2
Dodecylbenzenesulfonic acid	27176-87-0		CO
2-butoxyethanol	111-76-2		CA F2
2-butoxyethanol			
phosphoric acid	7664-38-2		CO

Legend

CA	Carcinogenic
CO	Corrosive
F2	Flammable - Second Degree
R2	Reactive - Second Degree

- Hazardous Substance List (Chapter 323) (PA-RTK)

Name acc. to inventory	CAS No	Classification
SULFURIC ACID	7664-93-9	E
BENZENESULFONIC ACID, DODECYL-	27176-87-0	E
ETHANOL, 2-BUTOXY-	111-76-2	
GLYCOL ETHERS		E
PHOSPHORIC ACID	7664-38-2	E

Legend

E	Environmental hazard

- Hazardous Substance List (RI-RTK)

Name of Substance	CAS No	References
Sulfuric acid	7664-93-9	T, F
2-butoxyethanol	111-76-2	T
2-butoxyethanol	111-76-2	T
phosphoric acid	7664-38-2	T, F
phosphoric acid	7664-38-2	T, F

Legend

F	Flammability (NFPA®)
T	Toxicity (ACGIH®)

Industry or Sector Specific Available Guidance(s)

NPCA-HMIS® III

Hazardous Materials Identification System. American Coatings Association.

Category	Rating
Chronic	*
Health	3
Flammability	0
Physical hazard	0
Personal protection	-

NFPA® 704

National Fire Protection Association: Standard System for the Identification of the Hazards of Materials for Emergency Response (United States).

Category	Degree of hazard
Flammability	0
Health	3
Instability	0
Special hazard	

National inventories

Country	Inventory	Status
US	TSCA	all ingredients are listed (ACTIVE)

Legend

TSCA Toxic Substance Control Act

SECTION 16: Other Information, Including Date of Preparation or Last Revision

OSHA Hazard Communication Standard (HCS), 29 CFR 1910.1200.

Classification Procedure

Physical and chemical properties: The classification is based on tested mixture.
Health hazards, Environmental hazards: The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

Disclaimer

This information is based on the present state of our knowledge and does not constitute an assurance of product properties nor establishes contract legal rights. All data about health and safety are only for information. They should therefore not be construed as specifications. This SDS has been compiled and is solely intended for this product.